DHINAKAR R



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PROFESSIONAL SUMMARY

Aspiring Electronics and Communication Engineering undergraduate aiming to secure a challenging position to Build an immersive career in a dynamic and growth-oriented organization, where I can utilize my knowledge, skills, and full potential to find innovative solutions for complex problematic scenarios.

ACADEMIC PROFILE

- ***B.E., Electronics and Communication Engineering*** **8.00**
  "K. S. Rangasamy College of Technology" **(**2023-2027**)** (Up to 6th Semester)
- ***HSE***
  "Reliance Matric Higher Secondary School" **(**2021-2023**)** **78.8 %**
- ***SSLC***
  "Government Higher Secondary School" **(**May-2021**)** **100%**

AREA OF INTEREST

- Integrated Circuits
- Semiconductor Devices
- Digital Electronics
- PLC

SKILL SET

- Coding Languages: C Programming, PLC Programming, Embedded Programming
- Simulation Tools: Multisim **|** Vivado **|** MATLAB **|** KiCad **|** WPLSoft

SOFT SKILLS

- Team Collaboration
- Active Listening
- Proactive Initiative
- Volunteer Leadership

PROJECTS

1. **Intelligent ATM Controller with security and transaction handling using VLSI**
   - Designed and implemented an ATM controller using Verilog HDL to perform secure banking operations including PIN authentication, cash withdrawal, balance inquiry, and transaction management.
   - Developed and verified the Finite State Machine (FSM) architecture through simulation using Xilinx Vivado/ModelSim, ensuring reliable and secure operation.

2. **Smart Boot for Mining Workers**
   - Developed a smart safety boot to improve the safety of mining workers by monitoring environmental conditions and worker movement.
   - Integrated Arduino, GPS, GSM, vibration, and IR sensors to detect hazards, track worker location, and transmit emergency alerts in real time.

3. **Smart Soil Monitoring Sensor System with Fe^{2+} Nutrient Detection**
   - Prototyped a low-cost, real-time smart sensor system for multi-parameter soil health and iron content analysis tailored for agricultural deployment.
   - Implemented signal conditioning circuitry to interface analog probe data with microcontrollers, ensuring accurate analog-to-digital conversion and drift compensation.

MINI PROJECTS

1. **2SC5200 amplifier**
   Made an AUDIO AMPLIFIER with a Versatile Transistor named 2SC5200
2. **Digital Dice**
   Digitalized the traditional DICE with help of two popularly known IC's named IC 555 & CD 4017
3. **Knock switch**
   Made a worthy Project to Activate the Loads from a portable place with IC 555
4. **DC-DC boost convertor**
   Did a Massive DC-DC BOOST CONVERTOR to use high range of Voltage needs from small source
5. **IR remote for appliances using TSOP1738**
   Did an optimization in one remote to control all loads within a circumstance
6. **PWM based speed control using IC 555**
   Make an PWM controlled circuit to gear up or down the DC motor

ACHIEVEMENTS

1. **MSME 5.0 –** Got Selected for Proposing the "Smart Badminton Kit and Racket" innovation.
2. **Smart India Hackathon (SIH) –** Idea got selected in National Contest for the "AI-Based Smart Safety Flow".
3. **MSME 6.0 –** Selected Innovation at inter-college level for Proposing a "Smart Soil Sensor for Soil Iron Content Detection, integrating with nutrient sensor in Kolli Hills".

CERTIFICATIONS & CO-CURRICULAR ACTIVITIES

Completed various courses on NPTEL platform **Jan 2025 – Apr 2026**

Internet of things 4.0 with "Elite" Jan – Apr 2025
Successfully completed "Privacy and security in online social media". Jul – Oct 2025
Completed "Human Computer Interaction" with "Elite + Gold" Jan – Apr 2026

Certified by MATLAB with various courses in Electronics domain **Dec 2024 – Jan 2025**

App building onramp
Wireless Communication Onramp
Machine Learning Onramp

INTERNSHIP **20th Jan - 20th Feb 2026**

Successfully completed a two-week internship at Eduskill Foundation on "**Embedded system developer (Online Internship course).**"
Outcome: I get the basics and can able to explain it to my friends and juniors.

PAPER PRESENTATIONS

Participated at "**Hindusthan institute of college and technology**" presented a PPT on the topic of "*Nanotechnology*" on 30th Apr 2025.
Participated at "**Hindustan institute of technology**" presented a PPT on the topic of "*Drone technology*" on 19th Feb 2025.

WORKSHOPS

Attended Workshop held on the topic of "*Drone technology*" at "**Kongu Engineering College**" on 27th Jan 2024.
Attended workshop held on "**Hindusthan Institute of College and Technology**" on the topic of "*AI automated Electronics devices*" on 13th Mar 2024.

PROJECT CONTEST

Participated at project expo conducted by the "**K. S. Rangasamy College of Technology**" on 2nd Nov 2024.
Participated at inter department project expo for academics Laboratory held at "**K. S. Rangasamy College of Technology**".

MEMBERSHIPS

Joined as a member in "**IETE**" (Institution of Electronics and Telecommunication Engineers)
Joined in a "**IEEE**" (Institute of Electrical and Electronics Engineers) Membership

EXTRACURRICULAR ACTIVITIES

Community services.
Volunteering and Organizing events in tech Expos
Active Participation in College Technical Activities